Waseem Raza, PhD

Senior Machine Learning Engineer | Generative Modeling & Production ML

Python | Deep Learning | GANs & Autoencoders | Data Pipelines | MLOps

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LinkedIn | GitHub | Google Scholar | Portfolio

Professional Summary

Senior-level Machine Learning Engineer and researcher with a PhD in Electrical and Computer Engineering and 7+ years of combined industry and research experience designing, training, validating, and operationalizing ML systems. Strong Python and deep-learning background spanning PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, NumPy, SciPy, Pandas, GANs, autoencoders, transfer learning, domain adaptation, reinforcement learning, CNN/LSTM models, anomaly detection, and predictive modeling. Experience combines research-quality experimentation with production-oriented engineering: reusable data/model pipelines, Git-based development, CI/CD, Docker/Kubernetes, cross-domain validation, cloud analytics, and stakeholder-driven deployment. Led 100+ controlled ML experiments, analyzed 500K+ operational logs, improved autoencoder-based compression by 25%, and delivered production-oriented ML workflows across AT&T, Futurewei, university research, and geoscience analytics.

Technical Skills

Generative Modeling & Representation Learning: GANs, autoencoders, deep neural networks, synthetic-data generation, transfer learning, domain adaptation, representation learning, transformer fine-tuning concepts, generative-model evaluation.
ML / Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Hugging Face Transformers; CNNs, LSTMs, reinforcement learning, classification, regression, anomaly detection, time-series modeling, model evaluation.
Python & Numerical Computing: Python, NumPy, SciPy, Pandas, SQL, C/C++, Jupyter, numerical experimentation, statistical analysis, vectorized processing, Linux, Bash/CLI.
Training, Inference & Performance: preprocessing, feature engineering, train/validation/test design, hyperparameter experimentation, cross-validation, batch inference, metric tracking, robustness testing, failure analysis, distribution-shift evaluation, model comparison.
Data Pipelines & Large-Scale Processing: Azure Databricks, Apache Spark, PySpark, Synapse Analytics, Azure Functions, ETL/ELT, structured/unstructured data, data-quality validation, AI-ready datasets, synthetic-data pipelines.
MLOps & Software Engineering: Git, GitHub Actions, CI/CD, Docker, Kubernetes, OpenShift, testing, REST APIs, Flask, Linux, reproducible repositories, deployment-oriented model validation, technical documentation.
Collaboration & Leadership: technical ownership, cross-functional requirements translation, research mentoring, experiment review, reproducibility standards, technical writing, presentations, and knowledge transfer.

Professional Experience

Machine Learning Engineer **02/2025 – Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Own end-to-end development, testing, and operational integration of ML diagnostics for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Design domain-adaptive neural models and reusable data/model pipelines that improve generalization across heterogeneous operational domains and distribution shifts.
- Develop multimodal AI prototypes combining text, location, network context, and operational signals; build error-analysis and failure-mode assessment procedures for deployment readiness.
- Partner with network engineers, software teams, and public-safety stakeholders to translate operational objectives into measurable ML requirements, validated datasets, model-selection decisions, and deployable workflows.

Data Scientist and AI/ML Engineer **09/2024 – 12/2024**
Impac Exploration Services *Weatherford, OK*

- Built ML and LLM-assisted analytical workflows for large geochemical datasets, collaborating with geologists to translate scientific objectives into scalable data and modeling pipelines.
- Developed ingestion, cleansing, feature-preparation, transformation, and validation workflows using Python, Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing, reducing data-preparation time by 40%; implemented cross-validation and data-quality checks that improved reliability of analytical outputs by 20%.

Graduate Research Assistant **09/2019 – 07/2025**
AI4Networks, University of Oklahoma *Tulsa, OK*

- Led research spanning generative modeling, robust ML, reinforcement learning, domain adaptation, wireless digital twins, network optimization, and wearable intelligence.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, CNN/LSTM, and reinforcement-learning pipelines

for sparse-data, cross-domain, and sequence-modeling problems.
- Designed reusable Python simulation/data-generation systems and robustness-evaluation frameworks; led 100+ controlled experiments from formulation through quantitative validation and publication.
- Mentored junior researchers in ML implementation, experiment design, reproducibility, technical writing, and presentation of results.

Advanced Wireless Technology Development Intern **09/2022 – 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to evaluate reconstruction performance, robustness, and generalization across changing data conditions; translated results into recommendations for next-generation communication systems.

Advanced-Degrees Intern **06/2024 – 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Developed and validated data-driven decision logic through structured experimentation, performance analysis, and communication of results to engineering teams.

Education

PhD, Electrical and Computer Engineering *09/2019 – 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 – 08/2016*
University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 – 06/2014*
University of Engineering and Technology, Taxila, Pakistan

Selected Machine Learning Projects

Domain-Informed GANs for Propagation Models & Digital Twins *Research Project | 2024*

Developed generative-modeling workflows for wireless digital twins using 3GPP-informed neural pretraining, GAN-based synthetic-data generation, and scarce measured data. Compared synthetic-data quality and downstream regression performance, and evaluated robustness under domain shift and limited real training data.
Tools & methods: Python, GANs, deep neural networks, synthetic data, domain adaptation, transfer learning, regression, RMSE/R^2, statistical validation, distribution-shift evaluation.

Autoencoder CSI Compression & Transfer Learning *Industry Research Project | 2022–2023*

Developed and evaluated autoencoder-based representation/compression workflows for large channel-state-information datasets. Applied transfer learning to improve performance by 25%, executed 50+ controlled experiments, and benchmarked reconstruction accuracy and generalization across changing data conditions.
Tools & methods: Python, autoencoders, deep learning, representation learning, transfer learning, signal compression, train/validation experimentation, performance benchmarking, robustness analysis.

Wearable Intelligence for Health Monitoring *Research Project | 2024–2026*

Developed deep-learning pipelines for medication-taking and dietary-activity recognition from smartwatch accelerometer/gyroscope data. Implemented segmentation, feature engineering, CNN/LSTM-style sequence models, cross-user validation, and recall-oriented error analysis for real-world health-monitoring scenarios.
Tools & methods: Python, CNN/LSTM, time-series modeling, sensor preprocessing, feature engineering, sequence classification, cross-user evaluation, confusion matrices, model comparison.

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020–2024*

Built a reusable Python simulation and synthetic-data environment for large-scale experiments where measured data is sparse or expensive. Generated controlled mobility, propagation, and network-state datasets to support repeatable model training, benchmarking, and robustness studies.
Tools & methods: Python, numerical simulation, synthetic data, experiment automation, reproducibility, 3GPP models, data pipelines, performance evaluation.

AI-Ready Lakehouse Pipeline *Portfolio Project | 2024*

Designed scalable ingestion, transformation, validation, and feature-preparation workflows for analytics and downstream ML using a cloud lakehouse architecture. Focused on repeatability, data quality, and scalable processing for model development and inference workflows.
Tools & methods: Azure Databricks, Apache Spark, PySpark, ETL/ELT, data ingestion, schema transformation, data validation, feature preparation, cloud data engineering.

Cloud Computer Vision *Guided Project | 2025*

Implemented image-classification, object-detection, and vehicle-counting workflows using managed cloud models and

Python inference pipelines. Validated preprocessing, prediction outputs, class/count aggregation, and model behavior across visual inputs.

Tools & methods: Python, TensorFlow concepts, computer vision, object detection, image classification, preprocessing, inference, model evaluation, Google Cloud AutoML.

Domain-Adaptive E911 Diagnostic ML System *Industry ML System | 2025–Present*

Developed and operationalized a machine-learning diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%. Built reusable preprocessing, domain-adaptive modeling, cross-domain evaluation, and failure-analysis workflows to support reliable deployment across heterogeneous operational conditions.

Tools & methods: Python, supervised classification, NLP/text features, domain adaptation, data preprocessing, cross-domain validation, error analysis, failure-mode assessment, production integration, stakeholder-driven evaluation.

Production ML, Reproducibility & Technical Leadership

- Build ML workflows as reusable stages for data preparation, model training/inference, metric computation, error analysis, and validation; document assumptions, experiments, and acceptance criteria.
- Use Git, automated tests, GitHub Actions, CI/CD, Docker, Kubernetes, and command-line workflows to support reproducibility and production-oriented engineering practices.
- Review model behavior for data leakage, metric misuse, shape/type errors, cross-domain failures, and mismatches between intended and actual model behavior before deployment decisions.
- Mentor junior researchers and collaborate with engineers, scientists, and domain experts to translate research objectives into reproducible implementations and measurable results.

Selected Technical Training

Machine Learning *Stanford Online / Coursera | Completed 03/2020*

Supervised/unsupervised learning, regression, classification, regularization, clustering, bias/variance, optimization concepts, and practical model evaluation.

Generative AI Engineering and Fine-Tuning Transformers *IBM / Coursera | Completed 08/2026*

Transformer architectures, pretrained models, model adaptation, fine-tuning strategies, tokenization concepts, training/inference tradeoffs, and LLM evaluation.

Introduction to Deep Learning & Neural Networks with Keras *IBM / Coursera | Completed 08/2026*

Neural-network architecture, activation functions, optimization, model training, validation, and iterative performance improvement.

Introduction to Computer Vision with TensorFlow *Google Cloud / Coursera | Completed 08/2024*

Image preprocessing, convolutional-network concepts, image classification, model construction, training, validation, and evaluation.

Python for Data Science, AI & Development *IBM / Coursera | Completed 07/2026*

Python programming, data structures, data manipulation, NumPy/Pandas-oriented analytical workflows, and reusable application development for data science and AI.

Selected Research Evidence

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *IEEE PIMRC, 2024*

Generative augmentation, neural modeling, scarce-data learning, robustness evaluation, and downstream predictive-performance validation.

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches *HealthINF, Accepted 2026*

Deep-learning sequence classification, multimodal sensor data, cross-user validation, and real-world reliability analysis.

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *IEEE SmartNets, 2024*

Machine-learning-based anomaly/outage detection, model evaluation, and decision support in challenging operational environments.

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *Sensors, 2025*

Predictive modeling, optimization, reliability analysis, and systematic evaluation for data-driven network operation.

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN *IEEE WCNC, 2022*

Data-driven characterization, optimization, and quantitative evaluation of positioning uncertainty and its impact on network decisions.